

Dynamic Resource Allocation for Indoor VLC/WiFi Heterogeneous Network Based on Q-learning

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Abstract—The rapid increase in user data traffic has led to conventional single radio frequency (RF) wireless communication no longer meeting user demand. Light-emitting diode (LED) visible light communication (VLC) can also not meet the communication demand due to its over-reliance on line-of-sight transmission. The heterogeneity of visible light networks and traditional RF networks can meet user demand well. However, because the communication link limits LED optical communication, the traditional resource allocation algorithm can no longer meet user demand and system utilisation. This paper addresses the issue by proposing a joint method for channel detection and access based on Q-learning multi-agent and a dynamic strategy for bandwidth adjustment for fixed channels. This approach resolves the issue of multiple users frequently switching links. It optimises the VLC communication constrained by the communication link using the benchmark the user transmits in the optimal channel link. Simulation results show that the proposed strategy dramatically improves the system's throughput, utilisation and user fairness compared with the traditional link-switching method with fixed bandwidth.

Keywords—visible light communication, resource allocation, heterogeneous networks, reinforcement learning, fairness

I. INTRODUCTION

In recent years, with the advent of the 5G era, internet data traffic has grown exponentially, and users' demands for higher data rates have intensified. However, traditional wireless radio frequency (RF) communication faces limitations regarding spectral performance, particularly in high hotspot environments with many people. Researchers have begun exploring alternative communication methods, such as visible light communication (VLC)[1]. VLC utilizes LED light sources as transmitters, but it is affected by the linear propagation of light[4]. In line-of-sight (LOS) scenarios, VLC performs well; however, it does not apply to non-line-of-sight links (such as communication through walls). Using VLC in isolation is greatly restricted. Therefore, a heterogeneous network structure combining VLC and RF communication has been derived to address this challenge[3]. This hybrid approach aims to meet the needs of high-speed data flow, expand transmission bandwidth, and solve the inherent problems associated with standalone VLC and RF communication[2].

Due to RF communication link attenuation in traditional communication networks, switching communication links at will to select an access point is possible. However, in heterogeneous networks, the communication link predominantly affects the visible link. Unlike the visible link, which occupies the bandwidth of the RF link, users cannot seamlessly switch to another VLC area within a VLC region. Therefore, detecting the communication link, optimizing channel selection, and refining the network structure becomes crucial.

In this paper, we design a link model for VLC, apply the reinforcement learning (RL) method to select the optimal channel and use a dynamic access point bandwidth policy based on the optimal channel. These enhancements significantly improve user fairness, satisfaction, and system throughput compared to the proportional fairness (PF) algorithm, maximum C/I (Max C/I), and round-robin (RR) algorithm.

II. METHOD

A. VLC Model

Fig.1 depicts the model for a single user linking to a VLC access point.

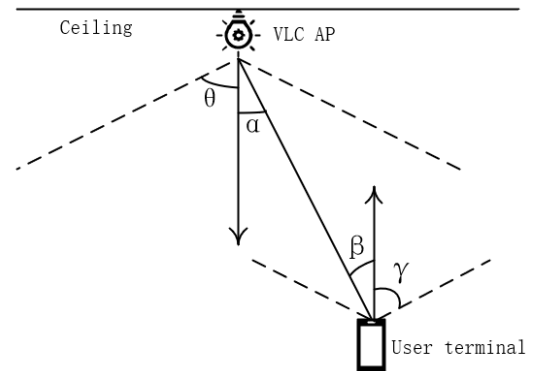


Fig. 1. Visible light communication model

In this model, the channel gain model for the user terminal to access the visible light access point is as follows:

$$T_{VLC}^{(i,j)} = \frac{(m+1)A_{pd} \cos^m(\alpha) \cos(\beta) G_{opt}(\beta) G_{c-opt}(\beta)}{2\pi(D^{(i,j)})^2} \quad (1)$$

In the formula(1):

$$G_{c-opt}(\beta) = \begin{cases} \frac{n^2}{\sin^2 \gamma} & 0 \leq \beta \leq \gamma \\ 0 & \beta > \gamma \end{cases} \quad (2)$$

n — Coefficient of refraction of optical lenses

γ — Elevated view of the terminal PD

m — Lambert order

θ — LED Half Power Angle

A_{pd} — Effective detection area of PD

α — LED radiation angle

β — Angle of incidence of the optical signal to the

PD

$G_{opt}(\beta)$ — Optical Filter Gain

$G_{c-opt}(\beta)$ — Optical Concentrator Gain

$D^{(i,j)}$ — Linear distance from the terminal to the access point

User terminal The SINR to access the visible light access point is as follows:

$$SINR_{VLC}^{(i,j)} = \frac{[\lambda T_{VLC}^{(i,j)} P_{VLC}]^2}{\sigma_{VLC}^2 b_{VLC}} \quad (3)$$

In the formula (3) :

λ — Photovoltaic conversion efficiency

P_{VLC} — Transmit power of the VLC access point

σ_{VLC}^2 — VLC channel noise power spectral density

b_{VLC} — VLC channel spectral width

The channel capacity of the user terminal to access the visible light access point is as follows:

$$v_{VLC}^{(i,j)} = \frac{1}{2} b_{VLC} \log_2(1 + SINR_{VLC}^{(i,j)}) \quad (4)$$

B. WiFi Model

User terminal The signal-to-noise ratio of the channel to the WIFI access point is as follows:

$$SINR_{WIFI}^{(i,j)} = \frac{P_{WIFI} G_{WIFI}^{(i,j)}}{b_{WIFI} \sigma_{WIFI}^2} \quad (5)$$

In the formula(5):

P_{WIFI} — Transmit power of WiFi

$G_{WIFI}^{(i,j)}$ — Channel gain from user i to AP j

b_{WIFI} — Channel Spectral Bandwidth for WiFi

σ_{WIFI}^2 — Power spectral density of WiFi

User terminal The channel capacity for accessing the WIFI access point is as follows:

$$v_{WIFI}^{(i,j)} = b_{WIFI} \log_2(1 + SINR_{WIFI}^{(i,j)}) \quad (6)$$

C. Q-learning Multi-Agent Channel Detection and Access

Assuming that there are (n-1) VLC access points and 1 WIFI access point (AP) hanging from the ceiling, where each VLC access point has (h) sub-channels, and the WIFI access point has (k) sub-channels. Additionally, there are (t) terminals scattered in different locations. Each terminal aims to access its locally optimal communication channel to achieve the maximum transmission rate. Meanwhile, the entire system seeks to maximize throughput. To address this problem, a reinforcement learning approach is employed for modeling[5],[6]. We propose a joint channel detection and access method based on a Q-learning multi-agent. The reinforcement learning process is shown in Fig.2.

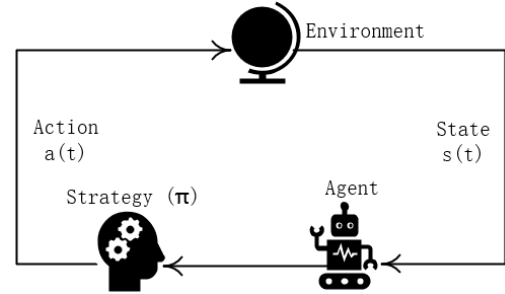


Fig. 2. Reinforcement of the learning process

The interaction process of reinforcement learning is a Markov decision-making process. The state at the current moment is only related to the state and action of the previous moment. At the current moment t, the environment gives a state s(t), the intelligent body chooses a strategy π to make an action a(t) based on this state, and at the same time, the environment gives a new state s(t+1) and reward feedback to the intelligent body. A reinforcement learning scheme for joint optimisation of multiple intelligences is designed in conjunction with the current environment

a. Agent: each user is an agent, and t users have t agents. Every agent wants to get the optimal quality of communication.

b. Environment: n-1 VLC APs and 1 WIFI AP, h(n-1) VLC channels and k WiFi channels.

c. Action space: each user can access the channels of different APs so there are (n-1) h+k actions.

d. State space: Assuming that the number of users is fixed, user access to the AP is a state, so there is a total of $\ln((n-1) h+k)$ states.

f. Reward: since each user wants to get the optimal channel that suits him/her, the maximum channel capacity of the next state s(t+1) - the channel capacity of the previous moment s(t) is the reward value, which is rewarded if it is positive,

neither rewarded nor penalised if it is 0, and penalised if it is negative. The reward equation is as follows:

$$r = v^{(i,j)} - v^{(i,k)} \quad (7)$$

In the formula (7): $v^{(i,j)}$ denotes the channel capacity of user i accessing to AP j , k is an AP different from j .

Each user wants to get the maximum channel capacity v used to ensure its own communication quality, thus it is considered to maximise the channel capacity for each user, so the problem can be formulated as:

$$\begin{aligned} \max \sum_{i=0}^{t-1} \sum_{j=0}^{n-1} a_{ij} V_{ij} \\ s.t. \ a_{ij} \in \{0,1\} \\ \sum_{j=0}^{n-1} a_{ij} = 1 \end{aligned} \quad (8)$$

In the formula (8): V_{ij} denotes the channel capacity that user i has access to AP j . a_{ij} denotes the access function of user i to AP j .

The steps of the algorithm for multi-agent training are shown in Fig 3.

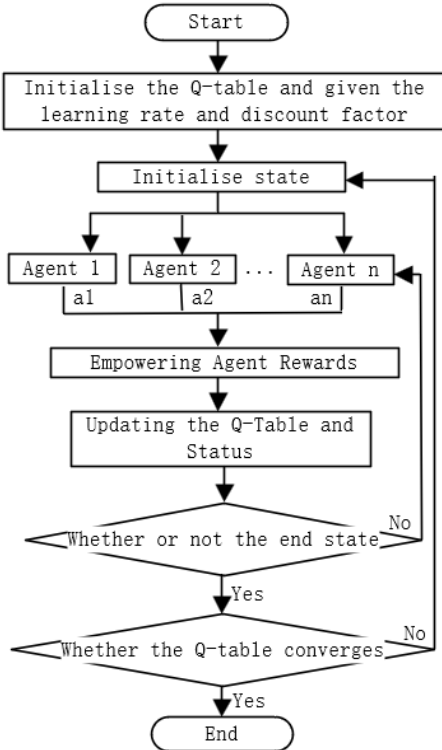


Fig. 3. Agent Training Process

D. Resource allocation algorithm

The current resource allocation algorithms are the RR algorithm, Max C/I algorithm and PF algorithm. RR algorithm pursues fairness by using a first-come-first-served strategy; Max

C/I pursues maximum bandwidth each time the bandwidth is the largest with the highest priority, which lacks fairness. PF algorithm provides a good compromise between fairness and bandwidth. The main ideas of the PF algorithm are as follows:

The PF algorithm uses a priority strategy to calculate the user's priority each time. The priority is calculated as follows:

$$p_i(t) = \frac{r_i(t)}{\bar{r}_i(t)} \quad (9)$$

In the formula (9):

$r_i(t)$ — The request rate of user t at the current moment

$\bar{r}_i(t)$ — The average request rate of user t at the current moment

$$\bar{r}_i(t) = \begin{cases} (1 - \frac{1}{t_c}) \bar{r}_i(t-1) + \frac{1}{t_c} r_i(t) & r_i(t) \neq 0 \\ (1 - \frac{1}{t_c}) \bar{r}_i(t-1) & r_i(t) = 0 \end{cases} \quad (10)$$

The priority of a user at any given moment is determined by the size of the current user's priority among all users.

E. Strategy for Dynamic Bandwidth Adjustment

Most current scheduling strategies adopt a fixed AP bandwidth allocation approach, where users select a relatively better AP for communication based on the current environmental state. While this method allows users to benefit from improved channel conditions across multiple APs, it has limitations in VLC. VLC communication has a narrow range, is typically effective within a 3-meter radius, and has better communication quality. Additionally, considering the role of illumination, only one VLC light source is available within this limited range. As a result, the heterogeneous network structure of VLC/Wi-Fi needs to be better suited for dynamically switching APs during communication. Therefore, the strategy of user-optimal communication channel and AP dynamic bandwidth adjustment can be an excellent solution to this problem as shown in Fig 4.

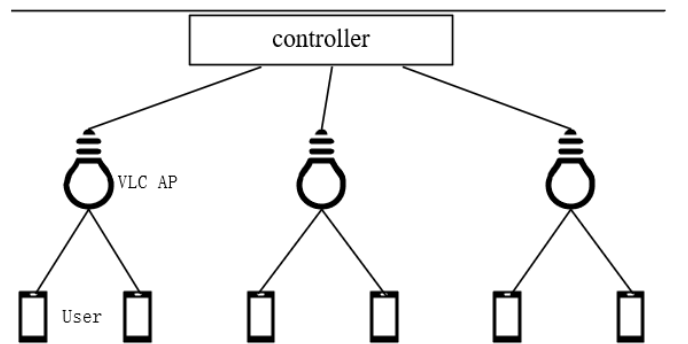


Fig. 4. Dynamic bandwidth adjustment system structure

The specific process is shown in Tab.1.

Tab. 1. AP dynamic bandwidth adjustment policy

- (1) At moment t all user terminals make a rate request to the access AP.
- (2) All APs receive the request without allocating the rate to the users first, and all APs send rate requests to the controller as the request speakers in their respective regions.
- (3) After the controller receives the requests from all APs, it adopts a policy to allocate the bandwidth to each AP at the moment t .
- (4) After obtaining the bandwidth at the current moment, each AP uses the PF, Max C/I and RR algorithms to allocate the AP's own bandwidth to the accessed users.
- (5) Repeat (1) (2) (3) (4).

Each AP must guarantee the real-time communication quality of the users, i.e., at least a specific user request rate must be obtained. At the same time, the bandwidth not used by other APs can also be obtained; the controller allocation policy adopts a dynamic proportional fairness algorithm. Each user enjoys a proportionality factor γ , and the acquired rate is γ multiplied by the total system bandwidth.

$$\gamma_i = \frac{R_i}{\sum_{i=0} R_i} \quad (11)$$

In the formula (11) :

R_i — The request rate of APi at time t .

The bandwidth obtained by APi is as follows:

$$B_i = \gamma_i B_{sys} \quad (12)$$

III. RESULT

The simulation is carried out in a 12mx12mx3m room with 9 VLC APs, 1 WIFI AP, and 30 users uniformly distributed throughout. A two-rectangular-dimensional coordinate system is established with the lower left corner of the room as the coordinate origin, and the coordinate system is depicted in Fig.5.

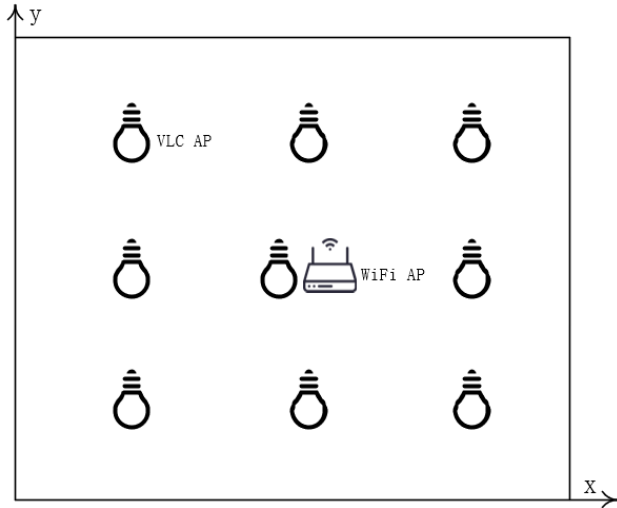


Fig. 5. AP Distribution Floor Plan

Each of the 30 users, who are dispersed across the area, is limited to one AP at a time. **Tab.2** illustrates the parameters for the system simulation.

Tab. 2. System simulation parameter

Parameter	Meaning	Value
n	Coefficient of refraction of a light lens	1.5
γ	Elevated view of the terminal PD	60°
α	LED radiation angle	60°
θ	LED Half Power Angle	60°
A_{pd}	Effective detection area of PD	1cm ²
$G_{opt}(\beta)$	Optical Filter Gain	1
λ	Photovoltaic conversion efficiency	0.53A/W
P_{VLC}	VLC transmit power	15W
σ_{VLC}^2	VLC channel noise power spectral density	-100 dBm/MHz
b_{VLC}	VLC sub-channel bandwidth	40MHz
P_{WIFI}	WiFi transmitter power	20dBm
b_{WIFI}	Channel bandwidth for WiFi	22MHz
σ_{WIFI}^2	Power spectral density of WIFI	-57 dBm/MHz
h	Number of individual VLC AP channels	10
k	Number of individual WIFI AP channels	10
B_{sys}	Total system bandwidth	1000Mbps

A. Channel capacity detection

Multiple users explore the system channel capacity and select APs using a multi-intelligence Q-learning approach, and the channel capacity profile is depicted in Fig.6.

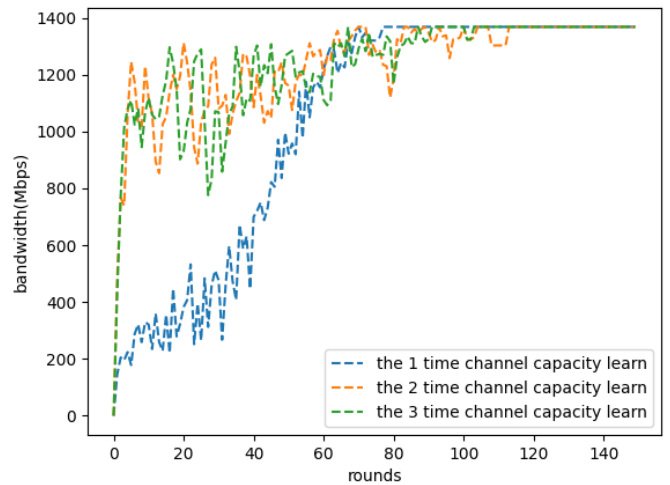


Fig. 6. System Maximum Capacity Detection

B. Dynamic Access Point Bandwidth Policy

Once the user has chosen a channel, simulate 100 rounds of random requests within that channel's capacity on each user terminal. Simulate the experiment on each AP using both fixed and dynamic bandwidth policies. The results of the AP simulation experiment are presented in Fig.7.

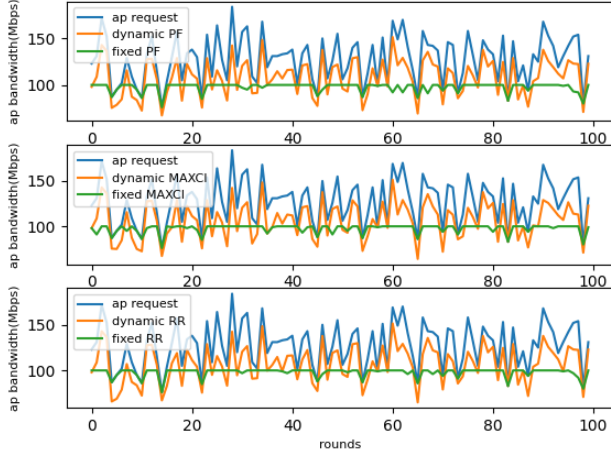


Fig. 7. AP Bandwidth Request Allocation

Fig.7 illustrates the allocation results using the three algorithms PF, MAX C/I and RR for equal requests, and it can be seen that the dynamic methods all outperform the fixed methods. The overall system bandwidth changes are shown in Fig.8.

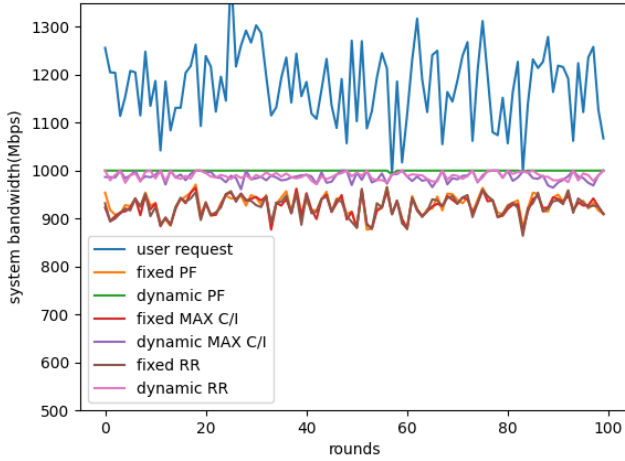


Fig. 8. Change in overall system bandwidth

As shown in Fig. 8, for all three benchmarking approaches, the dynamic AP bandwidth is larger than the fixed AP bandwidth. The most favourable results are achieved when the PF algorithm is combined with it.

C. User fairness index

Fig. 9 depicts the system user fairness index. In 100 rounds of simulation, the algorithm with dynamic PF's system fairness was the most stable and best among the three benchmark methods, with an average fairness index of 0.964.

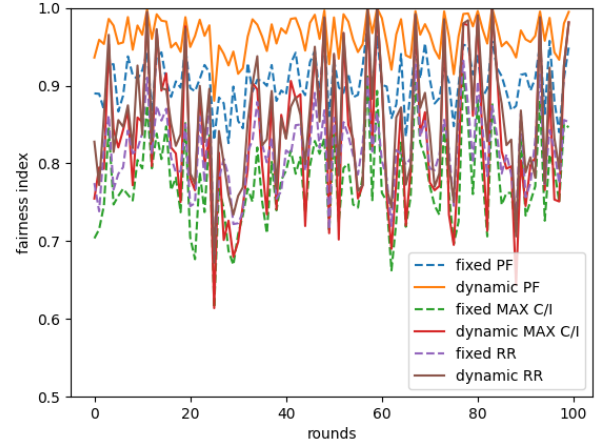


Fig. 9. System fairness index

IV. CONCLUSION

This paper analyses the channel characteristics of VLC/WIFI indoor heterogeneous networks for the new characteristics of channel dependence generated by VLC/WIFI. The dynamic bandwidth adjustment strategy is proposed based on the Q-learning multi-intelligent joint channel detection and access method and the fixed channel. It solves the problem of conventional multi-users switching the link frequently to satisfy their needs. It optimises the VLC subject to the constraints of the communication link. Simulation results show that 65.8 Mbps improves the overall average system throughput of the three primary methods, and the average user fairness index is improved by 0.04-0.06. The proposed model effectively improve the overall system throughput and user fairness.

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